

AIRFOIL DESIGN VIA CONTROL THEORY USING FULL POTENTIAL AND EULER EQUATIONS

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ABSTRACT

This paper describes an inverse method for airfoil design. The method employs control theory to make successive perturbations to a given airfoil such that a desired pressure distribution is obtained. The flow is modeled via both the full potential and the Euler equations. Several test cases are presented to demonstrate the effectiveness of the method for aerodynamic design.

INTRODUCTION

Until recently, airfoil design has largely been a process of trial and error that has relied heavily upon the intuition and experience of the designer. Using an experimental database for airfoils, a designer would normally employ an educated guess to make modifications to the design. He would then build a new model and test it in the wind tunnel. After several of these design iterations, an airfoil with the desired characteristics would be obtained.

Now, with the development of advanced Computational Fluid Dynamics (CFD) codes, much of the trial and error process in aerodynamic design can be eliminated. A CFD code models air flow around a given geometry by numerically solving the governing equations of fluid dynamics on the computer. Rather than testing a new model in the wind tunnel at each design iteration, it is much faster and cheaper to test new airfoil designs with CFD calculations. However each new design requires many (several hundred) computationally intensive CFD calculations before an acceptable design evolves using an optimization procedure. This process is called direct design optimization.

In order to alleviate the problems associated with the computationally intensive nature of the direct design methods, attempts have been made to calculate the shape of an airfoil that would have a specified flow distribution over its surface. This is essentially the inverse of a direct CFD calculation in which the shape of the airfoil is given and the flow is calculated; hence it is referred to as **inverse design**. Summaries of previous work in inverse design are given by Jameson [1], Cabuk and Modi [2], Lewis, Peters and Agarwal [3], and Agarwal and Lewis [4], and Lewis [5].

In this paper, the inverse design has been formulated as a variational problem in which a control system minimizes the differences between the actual and the desired velocity (or pressure distributions) over the airfoil surface by making successive changes to the shape of the airfoil. A flow chart of the design method is shown in Fig. 1. First, an initial airfoil and the desired velocity (or pressure distribution) are input. The Mach number and angle of attack are also specified. Then a flow solver (Full Potential or Euler) computes the actual velocity and pressure distribution around the airfoil. Next, a cost functional, which represents the deviation of the actual velocity (or pressure distribution) from the desired velocity (or pressure distribution) is computed. If the cost functional has reached a minimum, then the process terminates. Otherwise, a variation is made to the shape of the airfoil such that the cost functional is reduced. The process is repeated so that the cost functional is minimized thereby minimizing the deviation between the actual and the desired velocity (or pressure distributions). The inverse design enhances the efficiency of the aerodynamic design process by several orders of magnitude.

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Several test cases are shown to demonstrate the method and its robust characteristics (accuracy, stability, and efficiency) for aerodynamic design.

GOVERNING EQUATIONS

The flow field about the airfoil is modeled in the xy -plane by the full potential equation or the Euler equations subject to appropriate boundary conditions in a finite domain $\bar{D}$ around the airfoil as shown in Fig. 2.

Full Potential Equation

In Cartesian coordinate, the full potential equation can be written in terms of a velocity potential ϕ as

$$\frac{\partial}{\partial x} \left(\rho \frac{\partial \phi}{\partial x} \right) + \frac{\partial}{\partial y} \left(\rho \frac{\partial \phi}{\partial y} \right) = \nabla \cdot \rho \nabla \phi = 0 \quad (1)$$

Euler Equations

Euler equations can be written as

$$\frac{\partial w}{\partial t} + \frac{\partial \bar{F}}{\partial x} + \frac{\partial \bar{G}}{\partial y} = 0 \quad (2)$$

where

$$w = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}, \quad \bar{F} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho uH \end{bmatrix}, \quad \bar{G} = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ \rho vH \end{bmatrix},$$

$$p = (\gamma - 1) \rho \left[E - \frac{1}{2} (u^2 + v^2) \right], \text{ and } \rho H = \rho E + p.$$

COORDINATE TRANSFORMATION

In the inverse design method presented in this paper we employ an analytical mapping for successive perturbations of the airfoil although the method can be easily generalized to arbitrary grid perturbations.

In the present approach, the physical domain $\bar{D}$ is conformally mapped to an annular computational domain D as shown in Fig. 2.

Physical Domain

The airfoil is modeled by the curve $\bar{C}$ in the z -plane, where $z \equiv x + iy$. The nondimensional chord length is unity and the axes are placed such that the leading and trailing edge points are at $z = -0.5$ and $z = 0.5$ respectively. The flow field about the airfoil is modeled by Eq. (1) or (2) in the finite domain $\bar{D}$ around the airfoil subject to the inviscid flow

boundary condition. The far field boundary $\bar{B}$ is a circle centered at the origin with its radius $r_{\bar{B}}$ chosen to be large enough to approximate free stream conditions (usually several chord lengths).

Computational Domain

The computational domain consists of the annulus D defined in the σ -plane, where $\sigma \equiv re^{i\theta}$. The inner boundary C is chosen to be the unit circle centered at the origin. The outer boundary B is also a circle centered at the origin whose radius r_B depends on the geometry of the physical domain, as described in the next section.

Analytical Mapping

The subsequent development requires that the mapping between $\bar{D}$ and D be analytic.

$$\text{Define } \mu \equiv \frac{r_C}{r_B} = \frac{1}{r_B} \text{ and } \nu \equiv \frac{r_{\bar{B}}}{r_B}.$$

$$\text{Let } he^{i\beta} \equiv \frac{dz}{d\sigma} \text{ and } f \equiv \ln(h). \quad (3)$$

Note that since $z(\sigma)$ scales the circle B to the concentric circle $\bar{B}$, $h \approx \nu$ and $\beta \approx 0$ on B . Also note that since $z(\sigma)$ is analytic in D , if $\bar{B}$ and C are specified and the point $\sigma = 1$ is the required map to the trailing edge of the airfoil $z = 0.5$, then the only other thing required to determine the entire mapping is f on C .

Mapped Governing Equations

The governing equations may be written on the new annular domain D by transforming the coordinates (x, y) to (r, θ) . Given the above mapping function, we may now write the Jacobian of the coordinate transformation as

$$J = \frac{\partial(r, \theta)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{c}{h} & \frac{s}{h} \\ -\frac{s}{rh} & \frac{c}{rh} \end{vmatrix} = \frac{1}{rh^2},$$

where $s \equiv \sin(\theta + \beta)$ and $c \equiv \cos(\theta + \beta)$. Thus the full potential equation (1) becomes

$$r \frac{\partial}{\partial r} \left(r \rho \frac{\partial \phi}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\rho \frac{\partial \phi}{\partial \theta} \right) = \nabla \cdot \rho \nabla \phi = 0 \quad (4)$$

in the computational domain D .

The Euler equations (2) become

$$\frac{\partial(rh^2w)}{\partial t} + \frac{\partial(hF)}{\partial \theta} + \frac{\partial(rhG)}{\partial r} = 0, \quad (5)$$

where

$$w = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}, \quad F = \begin{bmatrix} \rho U \\ \rho uU - sp \\ \rho vU + cp \\ \rho UH \end{bmatrix}, \quad G = \begin{bmatrix} \rho V \\ \rho uV + cp \\ \rho vV + sp \\ \rho VH \end{bmatrix},$$

$$\text{and } \begin{bmatrix} U \\ V \end{bmatrix} = \begin{bmatrix} -s & c \\ c & s \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}.$$

Note that the flow tangency condition becomes $V = 0$ on C . The free stream velocity on the farfield boundary maps directly from $\bar{B}$ to B .

VARIATIONAL CALCULUS

The inverse method presented in this paper seeks to design an airfoil with a prescribed velocity (or pressure) distribution for a given Mach number and angle of attack. This task is cast as a problem in variational calculus where the velocity distribution is optimized by finding an optimal mapping function $z(\sigma)$. Throughout the design process, let $\bar{B}$ and C remain fixed and the point $\sigma = 1$ map to the trailing edge of the airfoil. As mentioned before the only other quantity required to determine the entire mapping is $f(\sigma)$ on C . Let $f(\theta)$ denote $f(\sigma)$ on C . Then for any $f(\theta)$, there is an associated airfoil $\bar{C}$ and an associated velocity distribution $q(\theta)$ or pressure distribution $p(\theta)$, which represents $q(z(\sigma))$ or $p(z(\sigma))$.

Full Potential Equation

In case of full potential equation, we assume that a desired velocity distribution $q_d(\theta)$ has been specified. We introduce a cost functional

$$I \equiv \frac{1}{2} \oint_C [q(\theta) - q_d(\theta)]^2 d\theta, \quad (6)$$

where $\oint$ represents a counterclockwise line integral, such that I represents the deviation of the actual velocity distribution $q(\theta)$ from the desired velocity distribution $q_d(\theta)$. The airfoil will be optimized when I is minimized; and the desired velocity distribution is obtained when $I = 0$.

Now consider a variation in the mapping function. Since the mapping is completely determined by

$f(\sigma)$ on C , the variation in the mapping is completely determined by δf on C . So the variation δf yields a variation in the shape of the airfoil which leads to a variation in the velocity distribution δq and ultimately to a variation in the cost functional 61.

To minimize I , the variation in the mapping function δf on C will be related to the variation in the cost functional 61. Then a perturbation in the mapping function will be chosen such that 61 is negative. Hence, when a new airfoil is generated from the perturbed mapping function, the resulting velocity distribution will be closer to the desired velocity distribution, causing I to decrease. Since any variation $\delta(\cdot)$ retains only first order terms, several iterations of the above procedure will be required to minimize I . Each iteration of the above procedure which results in a new mapping function and a new cost shall be referred to as a *design step*. The entire process used to minimize I which includes several design steps constitutes the design process.

We now begin the task of relating the variation in the cost functional 61 to the variation in the mapping function δf on C . First, the variation in the cost functional 61 is related to the variation in the velocity δq on C by

$$\delta I = \oint_C (q - q_d) \delta q d\theta. \quad (7)$$

Now since $q^2 = U^2 + V^2$,

$$\begin{aligned} \delta q &= \frac{U}{q} \delta U + \frac{V}{q} \delta V \\ &= \frac{U}{rhq} \frac{\partial(\delta\phi)}{\partial\theta} + \frac{V}{hq} \frac{\partial(\delta\phi)}{\partial r} - q\delta f \end{aligned} \quad (8)$$

But $V = 0$ and $r = 1$ on C . Therefore,

$$\delta q = \frac{U}{hq} \frac{\partial(\delta\phi)}{\partial\theta} - q\delta f \text{ on } C, \quad (9)$$

which may be substituted back into Eq. (7) to yield

$$\delta I = -\oint_C q(q - q_d) \delta f d\theta + \oint_C \frac{(q - q_d)U}{hq} \frac{\partial(\delta\phi)}{\partial\theta} d\theta. \quad (10)$$

The last integral in Eq. (10) may be integrated by parts to yield

$$\delta I = -\oint_C q(q - q_d) \delta f d\theta - \oint_C \delta\phi \frac{\partial}{\partial\theta} \left[\frac{(q - q_d)U}{hq} \right] d\theta. \quad (11)$$

Now, the variation of Eq. (4) yields the equation

$$L(\delta\phi) = -\nabla \cdot (\rho M^2 \delta f \nabla \phi), \quad (12)$$

where the operator L is defined as

$$L(\cdot) \equiv \nabla \cdot (\rho \nabla(\cdot)) - \nabla \cdot \left(\rho \left[\frac{\mathbf{q}}{c} \cdot \nabla(\cdot) \right] \frac{\mathbf{q}}{c} \right) \quad (13)$$

Thus Eq. (12) relates $\delta\phi$ to δf . We wish to use this relation to remove $\delta\phi$ from Eq. (11). To this end, a Lagrange multiplier ψ is introduced which satisfies

$$\begin{aligned} L(\psi) &= 0 & \text{in } D, \\ \psi &= 0 & \text{on } B, \end{aligned}$$

and
$$\frac{\partial \psi}{\partial r} = \frac{1}{\rho} \frac{\partial}{\partial \theta} \left[\frac{(q - q_d)U}{hq} \right] \quad \text{on } C. \quad (14)$$

Then the divergence theorem yields

$$\begin{aligned} \oint_C \delta\phi \frac{\partial}{\partial \theta} \left[\frac{(q - q_d)}{hq} \right] d\theta &= \oint_C \rho \delta\phi \frac{\partial \psi}{\partial r} d\theta \\ &= \int_D L(\psi \delta\phi) dA = - \int_D \psi L(\delta\phi) dA \end{aligned} \quad (15)$$

Now Eq. (12) and (15) may be used to eliminate $\delta\phi$ from Eq. (11) to yield

$$\delta I = - \oint_C q(q - q_d) \delta f d\theta - \int_D \psi \nabla \cdot (\rho M^2 \delta f \nabla \phi) dA \quad (16)$$

Thus we have δI in terms of δf . At this point, we would like to reduce the second integral to a line integral over C ; but first, we need to factor out δf . Note that

$$\oint_{\partial D} (\psi \rho M^2 \delta f \nabla \phi) \cdot \mathbf{n} dl = 0 \quad (17)$$

because $\psi = 0$ on B and $\nabla \phi \cdot \mathbf{n} = 0$ on C . So, by the divergence theorem,

$$\int_D \psi \nabla \cdot (\rho M^2 \delta f \nabla \phi) dA = - \int_D \rho M^2 \delta f \nabla \phi \cdot \nabla \psi dl \quad (18)$$

Now we introduce another auxiliary function P which satisfies

$$\nabla^2 P = \rho M^2 \nabla \phi \cdot \nabla \psi \quad \text{in } D \text{ and}$$

$$P = 0 \quad \text{on } B \text{ and } C. \quad (19)$$

By Green's second identity,

$$\begin{aligned} \int_D \nabla^2 P \delta f dA &= \int_D P \nabla^2 \delta f dA \\ &+ \oint_{\partial D} [\delta f (\nabla P \cdot \mathbf{n}) - P (\nabla(\delta f) \cdot \mathbf{n})] dl \end{aligned} \quad (20)$$

But $\nabla^2(\delta f) = \delta(\nabla^2 f) = 0$ in D because the mapping is analytic. Also, $\delta f = 0$ on B , and $P = 0$ on B and C . With these substitutions, Eq. (20) becomes

$$\int_D \nabla^2 P \delta f dA = \oint_C \frac{\partial P}{\partial r} \delta f d\theta \quad (21)$$

Substituting this back into Eq. (16) yields

$$\delta I = \oint_C \left[-q(q - q_d) + \frac{\partial P}{\partial r} \right] \delta f d\theta. \quad (22)$$

If we define

$$g \equiv -q(q - q_d) + \frac{\partial P}{\partial r} \quad (23)$$

and set

$$\delta f = -\lambda g, \quad (24)$$

where λ is a positive number used as a relaxation factor, then

$$\delta I = -\lambda \oint_C g^2 d\theta. \quad (25)$$

Note that $\delta I < 0$. This is desirable since I is positive and our goal is to make $I = 0$. Thus, if we compute a new mapping function which will result in a new airfoil by setting $f_{new} = f + \delta f$, then to a first order approximation, the actual change in the cost functional δI will be approximately equal to δI .

We compare I to δI in order to get a reasonable value for λ . Since we want $\delta I = -I$, we set

$$\lambda = \frac{I}{\oint_C g^2 d\theta} \quad (26)$$

so that $\delta I \approx -I$ and $I_{new} \approx 0$. However, this is only a first order approximation and λ may be adjusted in practice.

Euler Equations

In case of Euler equations, we assume that a desired pressure distribution $p_d(\theta)$ has been specified. We choose the cost function

$$I = -\frac{1}{2} \oint_C (p - p_d)^2 d\theta \quad (27)$$

The variation in the cost function δI is related to the variation in the pressure by δp on C by

$$\delta I = - \oint_C (p - p_d) \delta p d\theta. \quad (28)$$

The variation in the pressure δp is related to the variation mapping function δf through the variation in Eq. (5). Since Eq. (5) is solved to steady state,

$$\delta \frac{\partial(hF)}{\partial \theta} + \delta \frac{\partial(rhG)}{\partial r} = 0 \quad (29)$$

Equation (29) may be written in terms of δw instead of $6F$ and $6G$ by defining the Jacobian matrices

$$C = -s \frac{\partial F}{\partial w} + c \frac{\partial G}{\partial w} \quad \text{and} \quad (30)$$

$$D = c \frac{\partial F}{\partial w} + s \frac{\partial G}{\partial w}$$

and by noting that $\delta(hs) = s\delta h + hc\delta\beta$ and $\delta(hc) = c\delta h - hs\delta\beta$. Therefore

$$\delta(hF) = hC\delta w + F6h - hG\delta\beta \quad (31)$$

and

$$\delta(hG) = hD\delta w + G6h + hF\delta\beta \quad (32)$$

Thus Eq. (29) may be written as

$$0 = \frac{\partial}{\partial\theta}(hC\delta w) + \frac{\partial}{\partial r}(rhD\delta w) + \frac{\partial}{\partial\theta}(F\delta h - hG\delta\beta) + \frac{\partial}{\partial r}(rG\delta h - rhF\delta\beta). \quad (33)$$

Note that the flow tangency condition, $V = 0$ on C , is not affected by a change in the mapping function. Thus,

$$\delta V = 0 \quad \text{on } C. \quad (34)$$

Now, using Eq. (33) to "add zero" to Eq. (28) and introducing the Lagrange multiplier ψ , we have

$$\begin{aligned} \delta I = & - \oint_C (p - p_d) \delta p \, d\theta \\ & - \iint_D \psi^T \frac{\partial}{\partial r} (rhD\delta w + rG\delta h + rhF\delta\beta) \, dr \, d\theta \\ & - \iint_D \psi^T \frac{\partial}{\partial\theta} (hC\delta w + F\delta h - hG\delta\beta) \, dr \, d\theta \end{aligned} \quad (35)$$

which relates a variation in the cost function $6I$ to a variation in the mapping parameters $6h$ and $\delta\beta$.

Since the Lagrange multiplier ψ is arbitrary, it is now chosen such that Eq. (35) will be simplified. Let

$$\frac{\partial\psi}{\partial t} - C^T \frac{\partial\psi}{\partial\theta} - rD^T \frac{\partial\psi}{\partial r} = 0 \quad \text{in } D. \quad (36)$$

At the outer boundary B , we set characteristic boundary conditions such that there will be no incoming waves. In this case incoming waves correspond to positive eigenvalues of D^T . Let $D = TAT^{-1}$ where the columns of T are the eigenvectors of D , and let $\tilde{\psi}^+$ represent components of $\tilde{\psi}$ corresponding to positive eigenvalues of D^T . Then the far field boundary conditions may be written as

$$\tilde{\psi}^+ = 0 \quad \text{on } B. \quad (37)$$

On the body C , we set

$$h(c\psi_2 + s\psi_3) = p - p_d \quad (38)$$

We now use the adjoint equation (36) and its boundary conditions (37) and (38) to simplify Eq. (35). Applying Gauss's Theorem to second and third integrals in Eq. (35) and after considerable simplification we obtain [3],

$$\begin{aligned} \delta I = & \oint_C (p - p_d) p \delta f \, d\theta + \oint_C (c\psi_3 - s\psi_2) ph \delta\beta \, d\theta \\ & + \iint_D (F^T \frac{\partial\psi}{\partial\theta} + rG^T \frac{\partial\psi}{\partial r}) \delta h \, dr \, d\theta \\ & - \iint_D (G^T \frac{\partial\psi}{\partial\theta} + rF^T \frac{\partial\psi}{\partial r}) h \delta\beta \, dr \, d\theta. \end{aligned} \quad (39)$$

$$\text{Let } \nabla^2 P = \frac{h}{r} (F^T \frac{\partial\psi}{\partial\theta} + rG^T \frac{\partial\psi}{\partial r}) \quad (40)$$

$$\text{subject to } P = 0 \text{ on } B \text{ and } C, \quad (41)$$

$$\text{and } \nabla^2 Q = \frac{h}{r} (G^T \frac{\partial\psi}{\partial\theta} - rF^T \frac{\partial\psi}{\partial r}) \quad (42)$$

$$\text{subject to } Q = 0 \text{ on } B, \quad (43)$$

$$\text{and } \frac{\partial Q}{\partial r} = ph(s\psi_2 - c\psi_3) \quad \text{on } C. \quad (44)$$

Then using the Green's second identity on double integrals in Eq. (39), it can be shown [3] that,

$$\begin{aligned} \delta I = & \oint_C (p - p_d) p \delta f \, d\theta + \oint_C (c\psi_3 - s\psi_2) ph \delta\beta \, d\theta \\ & - \oint_C \frac{\partial P}{\partial r} \delta f \, d\theta + \oint_C \frac{\partial Q}{\partial r} \delta\beta \, d\theta - \oint_C Q \frac{\partial(\delta\beta)}{\partial r} \, d\theta \end{aligned} \quad (45)$$

Since $f + i\beta$ is to remain analytic in D , $\delta f + i\delta\beta$ also satisfies the Cauchy-Riemann equations. Thus

$$\frac{\partial(\delta f)}{\partial\theta} = - \frac{\partial(\delta\beta)}{\partial r} \quad \text{on } C. \quad (46)$$

Making this substitution and using the boundary condition (44), we can write

$$\delta I = \oint_C (p - p_d) p \delta f \, d\theta - \oint_C \frac{\partial P}{\partial r} \delta f \, d\theta + \oint_C Q \frac{\partial(\delta f)}{\partial\theta} \, d\theta. \quad (47)$$

which becomes

$$\delta I = \oint_C [p(p - p_d) - \frac{\partial P}{\partial r} - \frac{\partial Q}{\partial\theta}] \delta f \, d\theta. \quad (48)$$

$$\text{Let } g = p(p - p_d) - \frac{\partial P}{\partial r} - \frac{\partial Q}{\partial\theta} \quad (49)$$

$$\text{and } \delta f = \lambda g, \quad (50)$$

where λ is a positive number used as a relaxation factor, then

$$6I = \lambda \oint_C g^2 \, d\theta. \quad (51)$$

Note that $\delta I > 0$. This is desirable since I is negative and our goal is to make $I = 0$. Thus, if we compute a new mapping function which will result in a new airfoil by setting $f_{new} = f_{old} + \delta f$, then to a first order approximation, the actual change in the cost function $\mathbf{AI}$ will be equal to SI .

We compare I to δI in order to get a reasonable value for λ . Since we want $\mathbf{AI} = -I$, we set

$$\lambda = \frac{-I}{\oint g^2 d\theta} \quad (52)$$

so that $\mathbf{AI} \approx \delta I = -I$ and $I_{min} \approx 0$

AIRFOIL ADMISSIBILITY

The control theory provides no guarantee that the perturbed airfoil it produces can be physically constructed. For example, the new mapping function could map C to a curve that is not closed. Figure 3 shows two examples of this. For this reason, we set *admissibility criteria* on the shape of the curve $\bar{C}$; namely, $\bar{C}$ must be a simple closed curve.

Closure

Since the mapping function is analytic, once f_{new} has been determined on C by the control theory, the new values of dz/do are also determined on C . Thus the new airfoil $\bar{C}$ may be obtained by integrating dz/do over C via the formula

$$z(\sigma) = \int_1^\sigma \frac{dz}{d\sigma} d\sigma \quad (53)$$

In order that $\bar{C}$ be closed, the integration in Eq. (53) must start and end at the same location in the z -plane when the integral is evaluated over the closed path C . If we begin and end the integration at $\sigma = 1$, both endpoints will be at the trailing edge of the airfoil. Thus the distance between the starting and ending points of the integration is $|\Delta z|$ where

$$\Delta z = \oint_C \left(\frac{dz}{d\sigma} \right) d\sigma \quad (54)$$

So the airfoil will be closed when $\Delta z = 0$. To this end, we express $dz/d\sigma$ as the following Laurent series:

$$\frac{dz}{d\sigma} = \sum_{n=0}^{\infty} a_n \sigma^n + \sum_{n=1}^{\infty} a_{-n} \sigma^{-n} \quad (55)$$

where

$$a_n = \frac{1}{2\pi i} \oint_B \left(\frac{dz}{d\sigma} \right) \sigma^{-(n+1)} d\sigma \quad n = 0, 1, 2, \dots \quad (56)$$

and

$$a_{-n} = \frac{1}{2\pi i} \oint_B \left(\frac{dz}{d\sigma} \right) \sigma^{n-1} d\sigma \quad n = 1, 2, \dots \quad (57)$$

Note that

$$a_{-1} = \frac{1}{2\pi i} \oint_B \left(\frac{dz}{d\sigma} \right) d\sigma = \frac{\Delta z}{2\pi i} \quad (58)$$

Hence, the profile will be closed if $a_{-1} = 0$. So in order to insure closure, we may simply subtract $a_{-1}\sigma^{-1}$ from dz/do before integrating the perturbed airfoil. Figures 4 and 5 show the result of this.

Loop Removal

Even after insuring closure, there is still the possibility that $\bar{C}$ may cross over itself. However, since $\bar{C}$ starts out simple and the control theory makes only a small perturbation to it, only a small loop may occur near the endpoints of the integration of the new profile. When this happens, we simply replace the loop with two curve segments as shown in Fig. 6.

It is important that the transition between the substituted curve segments and the rest of the profile be smooth. For this reason parabolic curve segments are chosen such that they pass through both endpoints of the curve they replace and, at the left endpoint, they have the same slope as the curve they replace.

DESIGN PROCEDURE

The development in the previous sections is now reduced to the following procedure for the design of an airfoil using full potential (or Euler equations). The flow chart is shown in Fig. 1. The numerical methods used to carry out this procedure are described in the next section.

1. Choose an initial airfoil as a curve $\bar{C}$ defined by a set of points in the xy -plane. Also choose the desired velocity distribution q_d (or pressure distribution p_d).
2. Construct a grid around the airfoil by constructing an analytic mapping from points in the annulus D to points around the airfoil in $\bar{D}$.

3. Solve the governing equations (Eq. 4 or 5) on the grid constructed in step 2.
4. Compute the cost functional I via Eq. (6) or (27) and check to see if it is within a specified tolerance above zero. If it is, then we are done. If not, then continue with step 5.
5. Solve the adjoint equation (14) or (36) for ψ in D .
6. Solve the auxiliary Poisson equation Eq. (19) or (40) for P in D . For Euler case, also solve the Poisson equation (42).
7. Calculate g on C via Eq. (23) or (49).
8. Calculate λ via Eq. (26) or (52).
9. Calculate δf on C via Eq. (24) or (50).
10. Set $\delta\beta$ equal to the conjugate periodic function of δf on C as described in the next section.
11. Update f and β on C by setting $f_{new} = f + \delta f$ and $\beta_{new} = \beta + \delta\beta$.
12. Calculate a new dz/do on C via Eq. (3).
13. Calculate a_{-1} via Eq. (58) and subtract $a_{-1}\sigma^{-1}$ from the new dz/do on C .
14. Integrate a new airfoil by integrating $dz/d\sigma$ over C starting at the trailing edge. (If step 13 was done correctly, this integration will end at its starting point.)
15. If there is a loop at the trailing edge, remove it with a curve fit as described in the previous section.
16. Go to step 2.

NUMERICAL METHODS

In this section, we describe the numerical methods used to model the geometry and implement the design procedure.

Analytic Mapping and Grid Generation

The task of grid generation about the airfoil in physical domain is accomplished by the conformal mapping technique described in Halsey [6]. Since the mapping is analytic, x and y satisfy the Cauchy-Riemann equations and therefore Laplace's equation in D . Since $x(\sigma)$ and $y(\sigma)$ are known on B and C , a Laplace solver is used to fill the interior values and thus generating the grid in $\bar{D}$. Thus, $x(\sigma)$ and $y(\sigma)$ are specified on all the points in D . Since $z = x + iy$, the mapping function $z(\sigma)$ is determined when x and y are specified as functions of σ .

The derivative of the mapping function, dz/do , is approximated by

$$\frac{dz}{do} \approx \frac{\Delta z}{\Delta \sigma} \quad (59)$$

between grid points. Eq. (3) is then used to calculate f and β .

Flow Solver

Every attempt has been made to keep the inverse code independent of the flow solver. For obtaining the solution of the full potential equation, the code of Chen and Caughey [7] was used and for the solution of the Euler equations, the Euler code of Agarwal and Deese [8] was employed.

Adjoint Equation

The adjoint equation (14) can be written in the form

$$a \frac{\partial^2 \psi}{\partial r^2} + b \frac{\partial^2 \psi}{\partial r \partial \theta} + c \frac{\partial^2 \psi}{\partial \theta^2} + d \frac{\partial \psi}{\partial \theta} + e \frac{\partial \psi}{\partial r} + f \psi = g, \quad (60)$$

with $b^2 - 4ac = 4\rho(M^2 - 1)$. This equation is elliptic for subsonic flow and hyperbolic for supersonic flow. Eq. (60) is solved implicitly by writing it in discretized form to yield a system of linear equations of the form $[A]\mathbf{x} = \mathbf{b}$, where $[A]$ is a banded matrix. This system is then solved by successive over relaxation.

Solution of the Laplace and Poisson Equations

Step 2 in the design procedure requires solution of the Laplace equation and step 6 requires solution of the Poisson equation. The same routines are used for both the Laplace equation and the Poisson equation. The Poisson equation is solved implicitly by calculating a coefficient matrix for a linear system of equations of the form $[A]\mathbf{x} = \mathbf{b}$, and solving them with FISHPACK [9].

Conjugate Periodic Function

In step 10 of the design procedure, the calculation of $\delta\beta$ which is the conjugate periodic function of δf is required. We express δf as

$$\delta f(\theta) = \sum_{k=-m}^m a_k e^{ik\theta}, \quad (61)$$

so that $\delta\beta$ may be expressed as

$$\delta\beta(\theta) = \sum_{k=-m}^m b_k e^{ik\theta},$$

$$b_k = \begin{cases} ia_k & \text{if } k < 0 \\ 0 & \text{if } k = 0 \\ -ia_k & \text{if } k > 0 \end{cases} \quad (62)$$

Thus, given δf on C , the coefficients a_k are computed via an FFT. The coefficients b_k are then easily computed and passed to an inverse FFT to yield $\delta\beta$ on C .

Trailing Edge Loop Removal

Step 15 requires the removal of small loops near the tail of the airfoil when they occur. Our method is to replace the loop with two parabolic curve segments. Figure 6 shows a plot of a typical trailing edge before and after the loop removal.

The first step is to determine how much of the tail is to be replaced. We choose a point on the upper surface of the airfoil which is a few points to the left of the intersection of the two surfaces. This point is the left endpoint of the section of the upper surface to be replaced. The right endpoint is the point at the trailing edge. The same thing is done for the lower surface.

A parabolic curve is then defined such that it passes through the endpoints of the curve to be replaced and has the same slope as the original curve at the left endpoint. Then the points on the curve to be replaced are simply shifted vertically so that they lie on the curve. Again, the same thing is done for the section of the lower surface that is to be replaced.

RESULTS

Calculations based on Full-Potential Equations

A computer code has been written for inverse airfoil design implementing the procedure shown in Fig. 1 and described in Ref. [5]. In this section we present results of some calculations produced by this code.

Only a small percentage of the total CPU time required to design a new airfoil is spent computing perturbations to the profile. The rest is spent computing flow solutions. All of the test cases presented below were run on 0-grids containing 128 cells in the azimuthal direction and 64 cells in the

radial direction and used 11.2 hours of CPU time on a Silicon Graphics Indigo workstation with an R3000 CPU running at 33 MHz. Of that time, 8.3 hours of CPU time were spent in the flow solver, leaving only 2.9 hours for the design and grid generation.

(a) From a NACA 0006 to a NACA 0012 Airfoil

For the first test case, the initial airfoil was a NACA 0006 and the desired velocity distribution came from a flow solution run on a NACA 00012 airfoil. Thus both the initial and desired airfoils were known priori but only the initial airfoil was input to the inverse code. The goal then was for the design code to produce a NACA 0012 airfoil given only its velocity distribution. The evolution of both the airfoils and the velocity distributions are shown in Figs. 7 and 8. There is good agreement between the target and designed velocity distributions and the target and designed airfoils.

(b) From a NACA 0012 to a NACA 0006 Airfoil

The second test case is similar to the first. The initial airfoil was a NACA 0012 and the desired velocity distribution came from a NACA 0006 airfoil. The evolution of both the airfoils and the velocity distributions is shown in Figs. 9 and 10. Once again there is good agreement between the target and desired velocity distributions as well as the airfoils.

(c) From a NACA 0012 to a RAE 2822 Airfoil

The final test case was to go from a NACA 0012 to a RAE 2822 airfoil. The evolution of the airfoils and the velocity distributions is shown in Figs. 11 and 12. In this more difficult case, the inverse design method achieved fairly good agreement over most of the airfoil surface, but there is discrepancy in the target and desired velocity distributions as well as the airfoils near the tail. It is believed that more accurate numerical solution methods for the adjoint equation can resolve this difficulty. However, an extended variational formulation may be required which takes into account the second variation as well as the first variation. The problem may also be due to inaccuracies in the conformal mapping.

Calculation Based on Euler Equations

A second computer code has been written for inverse airfoil design based on Euler equations implementing the procedure shown in Fig. 1 and described in Ref. [3]. In this section we present results of some

calculations produced by this code. All the test cases that were run using the Full-Potential equations were repeated using the same grids.

(a) From a NACA 0006 to a NACA 0012 Airfoil

For this test case, as before, the initial airfoil was a NACA 0006 and the desired pressure distribution came from a flow solution run on a NACA 0012 airfoil. It should be noted that in this case desired pressure distribution is specified in the inverse design formulation based on the full-potential equation. Again both the initial and desired airfoils are known a priori but only the initial airfoil is input to the code. The goal then is for the code to produce a NACA 0012 airfoil given only its pressure distribution. Figures 13 and 14 show excellent agreement between the target and designed airfoils and pressure distributions.

(b) From a NACA 0012 to a NACA 0006 Airfoil

In this case the initial airfoil was a NACA 0012 and the desired pressure distribution came from a NACA 0006 airfoil. Again both the target and designed pressure distributions are in good agreement as shown in Figures 15 and 16.

CONCLUSIONS

In conclusion, two computer codes which use control theory to design airfoils using the Full-Potential and Euler equations to model the flow have been written with promising results.

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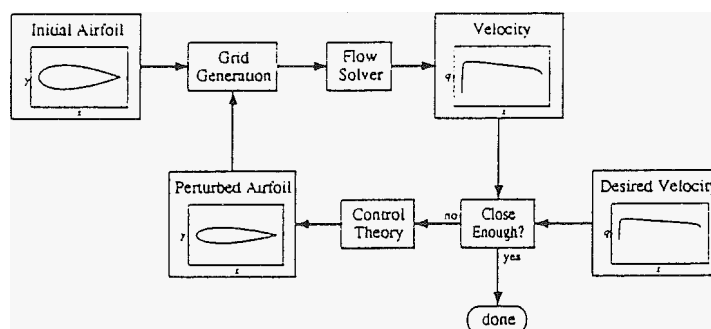


Figure 1. Flow chart of the design procedure.

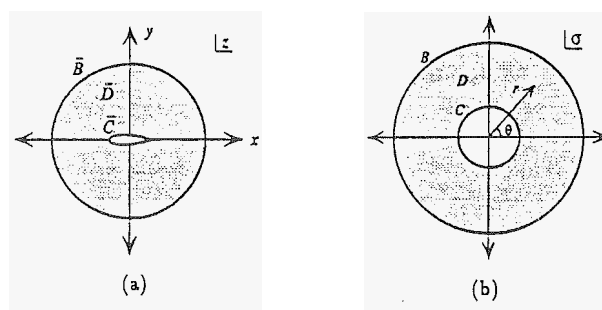


Figure 2. Coordinate transformation: a) Physical Domain, b) Computational Domain.

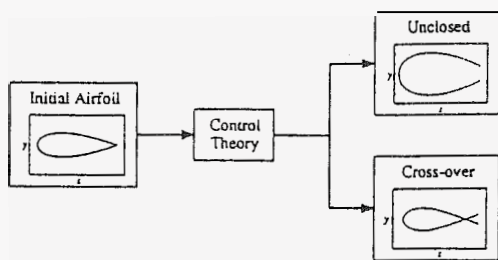


Figure 3. Unclosed profiles generated by the control theory.

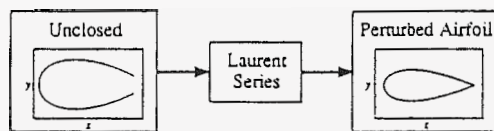


Figure 4. Use of a Laurent series to close a profile.

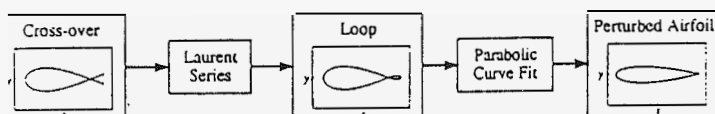


Figure 5. Use of a Laurent series to close a profile followed by a curve fit to remove a loop at the trailing edge.

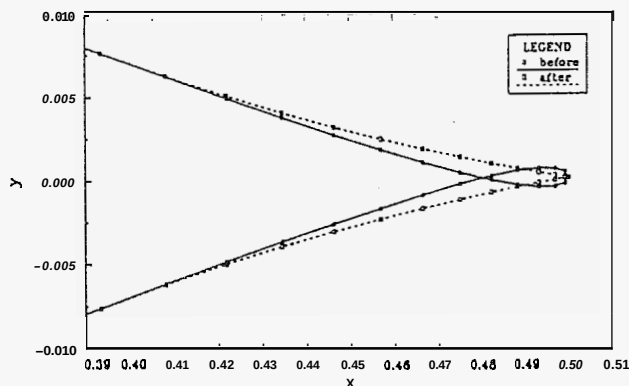


Figure 6. Removal of a loop at the trailing edge.

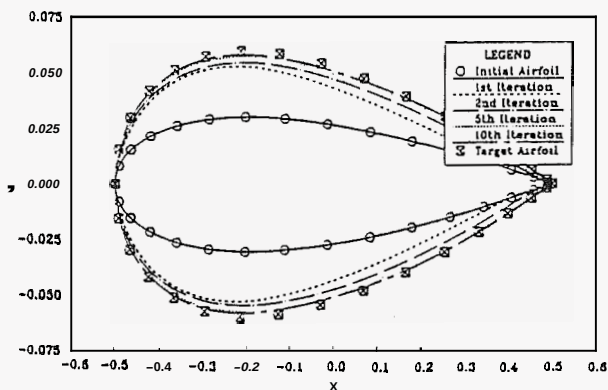


Figure 7. Airfoil evolution: The initial airfoil is a NACA 0006, and the desired velocity distribution is that computed on a NACA 0012 airfoil.

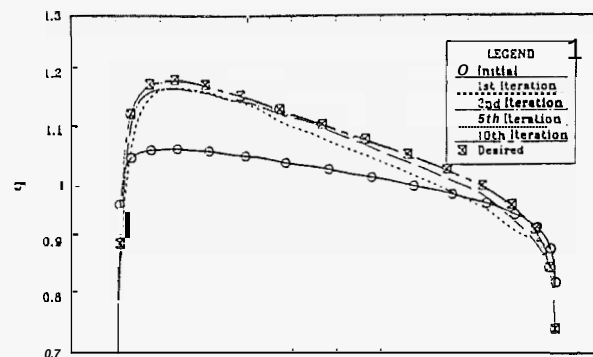


Figure 8. Evolution of the velocity distribution: The initial airfoil is a NACA 0006, and the desired velocity distribution is that computed on a NACA 0012 airfoil.

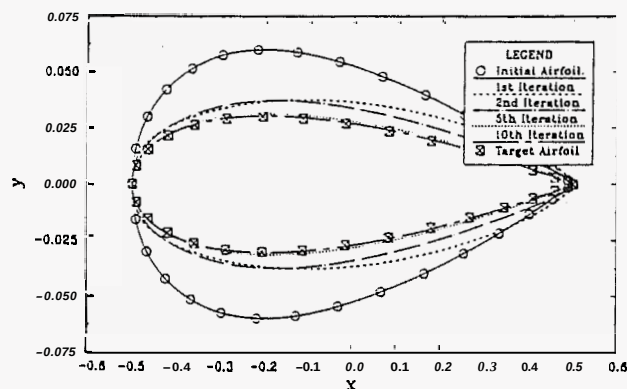


Figure 9. Airfoil evolution: The initial airfoil is a NACA 0012, and the desired velocity distribution is that computed on a NACA 0006 airfoil.

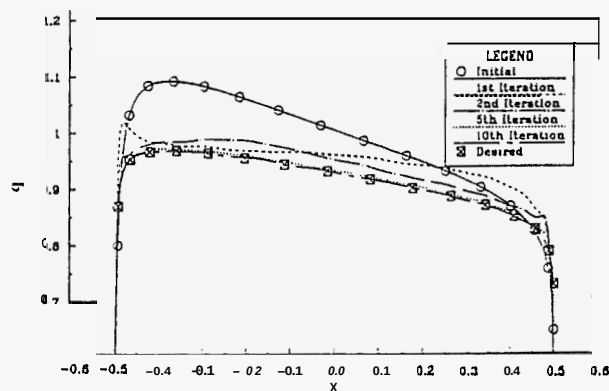


Figure 10. Evolution of velocity distribution: The initial airfoil is a NACA 0012, and the desired velocity distribution is that computed on a NACA 0006 airfoil.

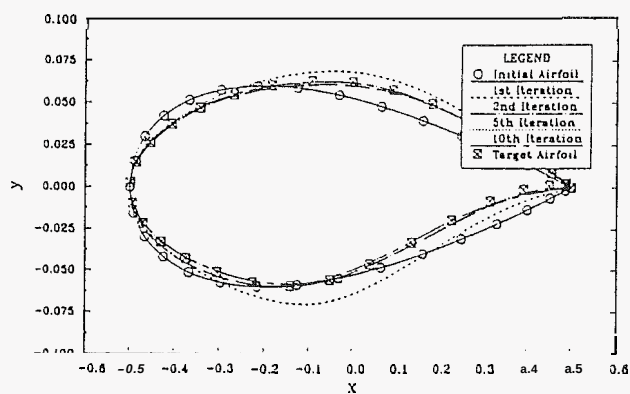


Figure 11. Airfoil evolution: The initial airfoil is a NACA 0012, and the desired velocity distribution is that computed on a RAE 2822 airfoil.

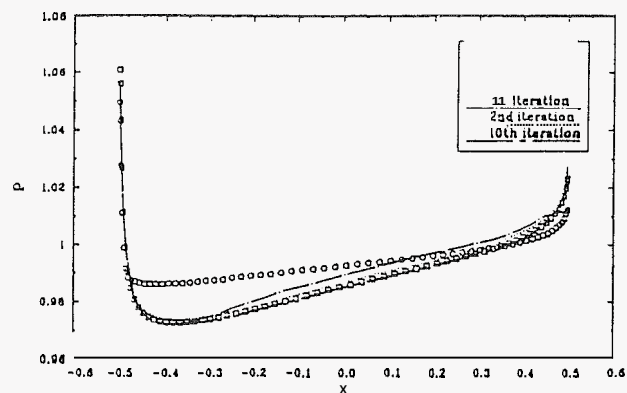


Figure 14. Pressure Evolution: The initial airfoil is a NACA 0006, and the desired pressure distribution is from a NACA0012 airfoil.

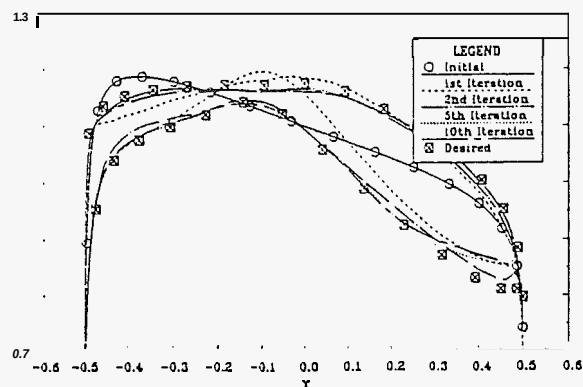


Figure 12. Evolution of the velocity distribution: The initial airfoil is a NACA 0012, and the desired velocity distribution is that computed on a RAE 2822 airfoil.

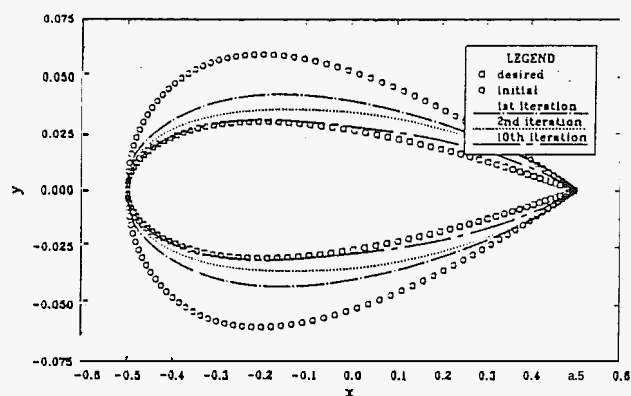


Figure 15. Airfoil Evolution: The initial airfoil is a NACA 0012, and the desired pressure distribution is from a NACA 0006 airfoil.

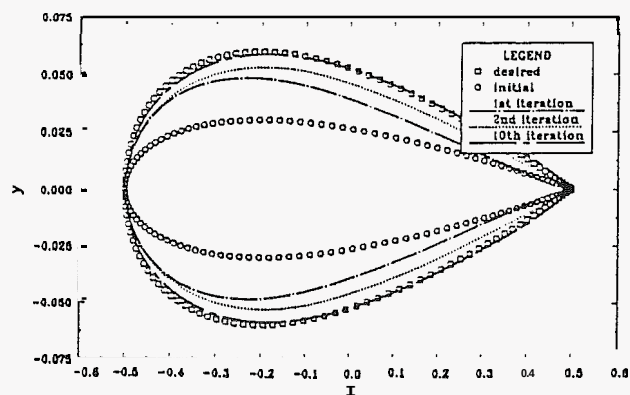


Figure 13. Airfoil Evolution: The initial airfoil is a NACA 0006, and the desired pressure distribution is from a NACA0012 airfoil.

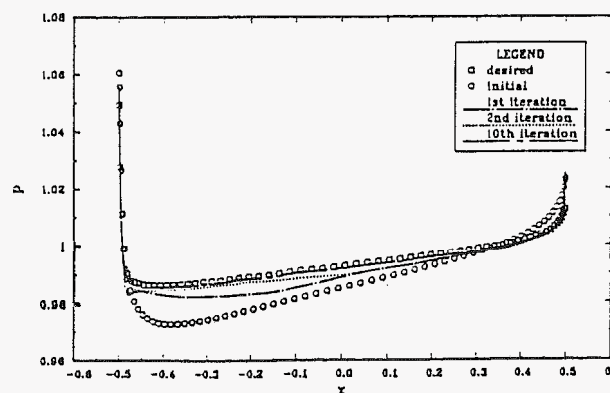


Figure 16. Pressure Evolution: The initial airfoil is a NACA 0012, and the desired pressure distribution is from a NACA0006 airfoil.